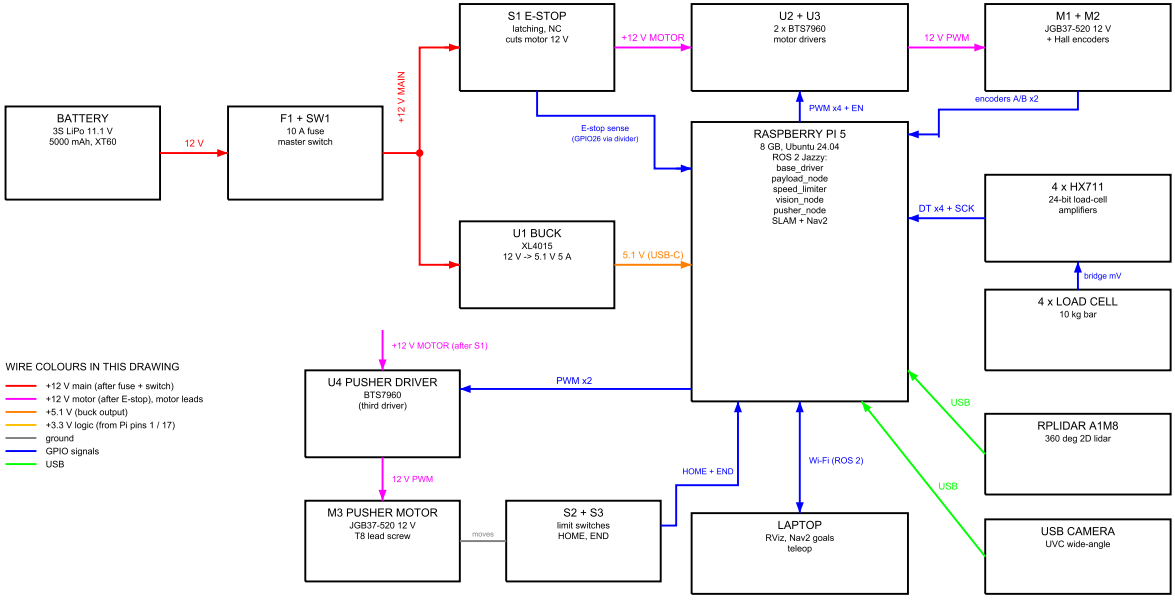
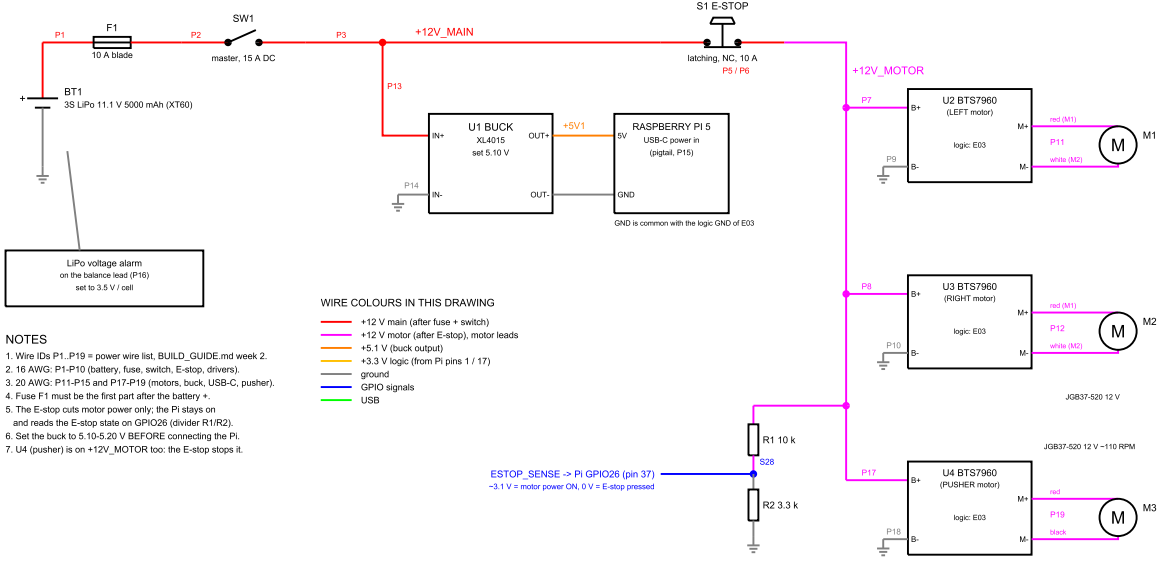


Details: E02 power distribution, E03 control & sensor wiring. Pin numbers: cargo_bot/pins.py.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: SYSTEM BLOCK DIAGRAM					DWG No.: CR-V3-E01
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 1 / 3	UNITS: -	TYPE: SCHEMATIC	DATE: 13/10/2026	DRAWN:



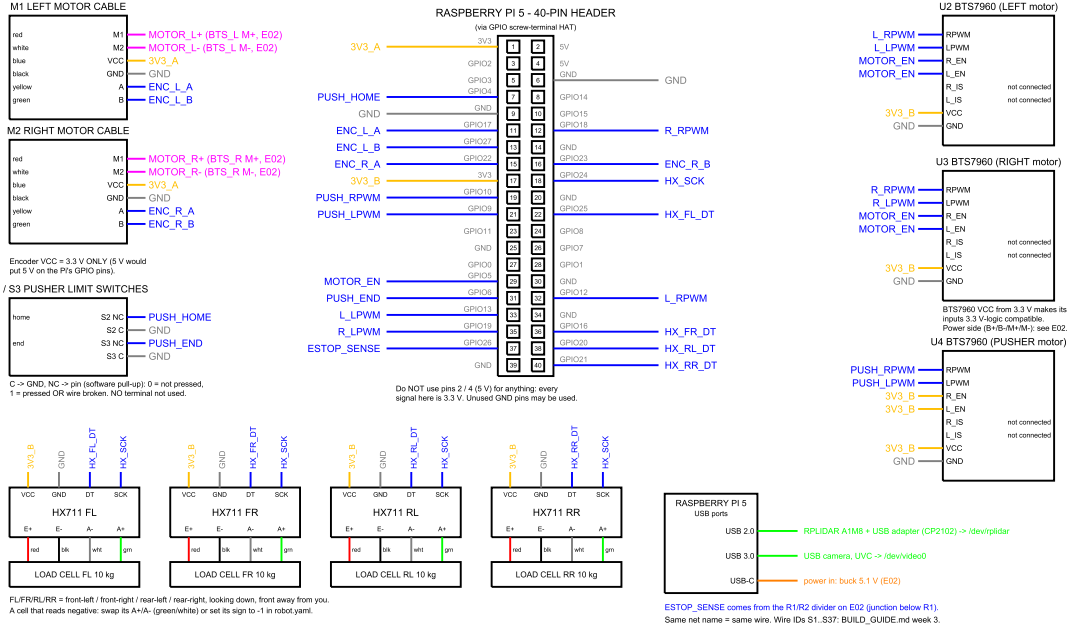
WIRE COLOURS IN THIS DRAWING

- +12 V main (after fuse + switch)
- +12 V motor (after E-stop), motor leads
- +5.1 V (buck output)
- +3.3 V logic (from Pi pins 1 / 17)
- ground
- GPIO signals
- USB

NOTES

1. Wire IDs P1..P19 = power wire list, BUILD_GUIDE.md week 2.
2. 16 AWG: P1-P10 (battery, fuse, switch, E-stop, drivers).
3. 20 AWG: P11-P15 and P17-P19 (motors, buck, USB-C, pusher).
4. Fuse F1 must be the first part after the battery +.
5. The E-stop cuts motor power only; the Pi stays on and reads the E-stop state on GPIO26 (divider R1/R2).
6. Set the buck to 5.10-5.20 V BEFORE connecting the Pi.
7. U4 (pusher) is on +12V_MOTOR too: the E-stop stops it.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: POWER DISTRIBUTION					DWG No.: CR-V3-E02
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: NTS	SHEET: 2 / 3	UNITS: -	TYPE: SCHEMATIC	DATE: 13/10/2026	DRAWN:



PROJECT	AUTONOMOUS CARGO ROBOT - DESIGN v3				
TITLE	CONTROL & SENSOR WIRING				DWG No. CR-V3-E03
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE	SHEET	UNITS	TYPE	DATE	DRAWN
NTS	3 / 3	-	SCHEMATIC	13/10/2026	